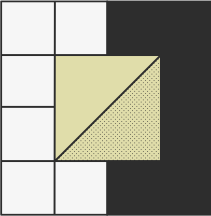


ONE NATION ONE HEALTHCARD

GROUP 4

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Harshanbarath RS	20BCE0954
Rujula Nitin Patil	20BCE2789
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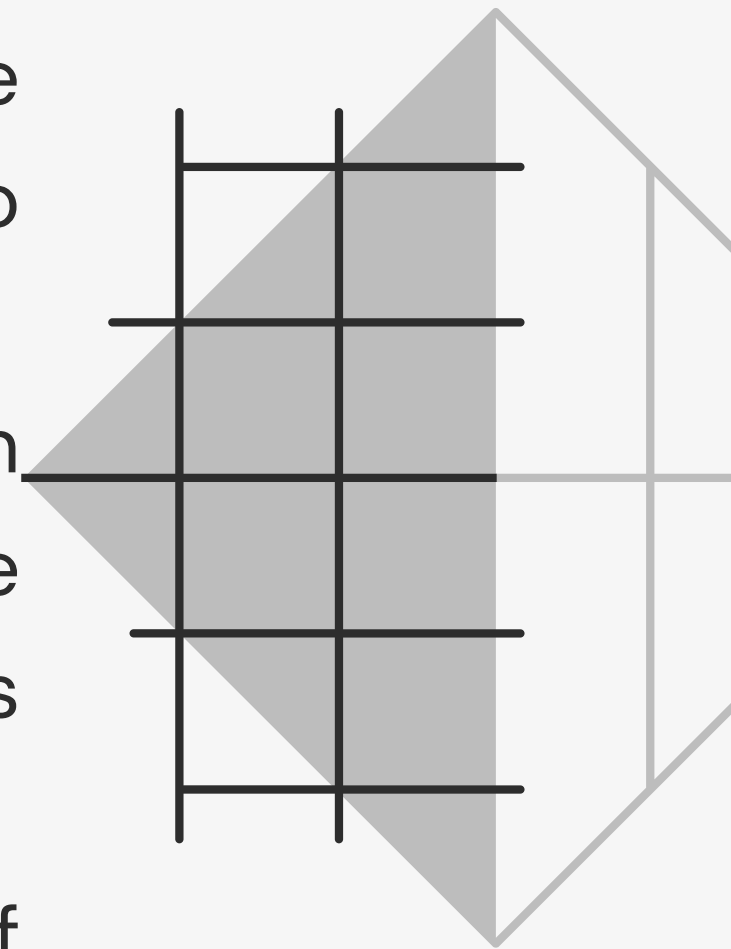


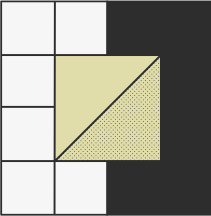
Problem Statement

Patients **face a lot of hassle** carrying their previous prescription every time they visit the doctor. This is also a hassle for the **doctors** as they must go through a lot of prescriptions in a day.

There simultaneously exists the problem where patients who have taken certain drugs in the past are prescribed drugs by another doctor. These interaction between the two drugs may **cause them harm** as the doctors may **not be aware of previous prescriptions**.

Further more making prescriptions everyday leads to the usage of a lot of paper which leads to the **cutting of trees** and other environmental damage.





Scope Statement

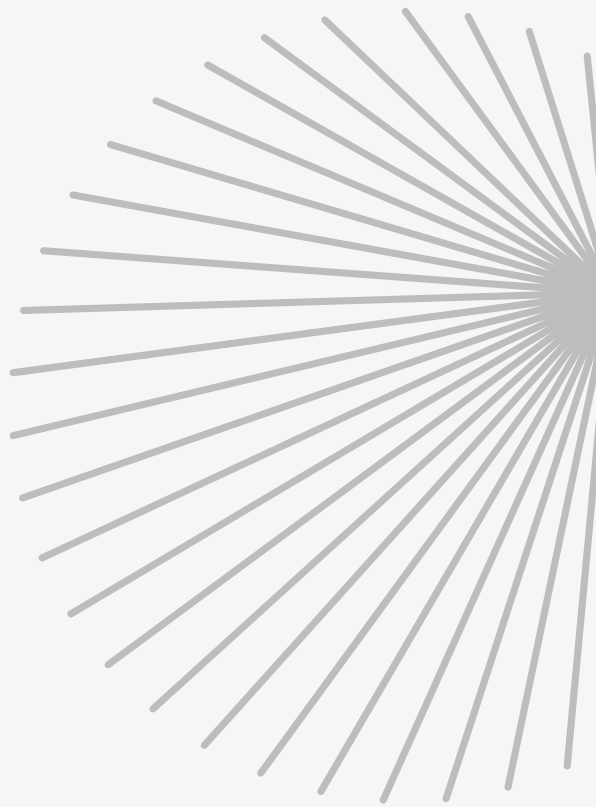
AIM:

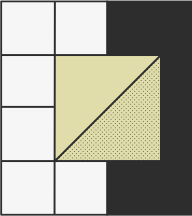


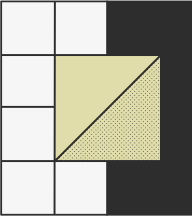
To develop a software that creates a **nationwide synced pharmaceutical system** to enable doctors, patients and pharmacist to create and track prescriptions of patients across India.

Objective:

- Efficiency
- Reliability
- Centralized



- 
- **Efficiency:** The system makes the doctor's appointment quite efficient since the patient **need not carry** any documents or store them.
 - **Reliability:** the system ensures that the data(prescriptions) that doctors get is **authentic and not tampered** with.
 - **Centralized:** As the system is used by patients across the country, all their data (history of the previous prescription) is saved and synced on a **single non-destructible server**.

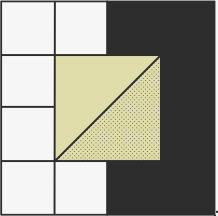


Process Model

The process model we have selected is the **Waterfall Model** as the project we are working on is a **small-scale project** which is **easy to implement**.

We are also **aware of all the requirements** before hand and know that there are **no possible changes** in requirements with regards to hospitals and users.

Waterfall model also requires that each stage have **detailed documentation** which is required by this project.



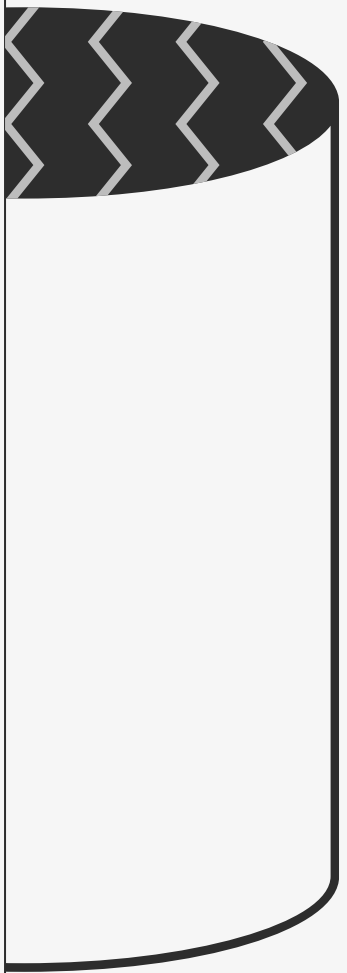
Stake Holders

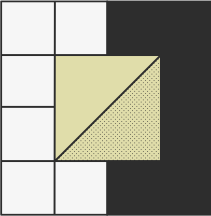
Patients

Doctors

Pharmacists

Admin





Requirements w.r.t Stake Holders

Patients:

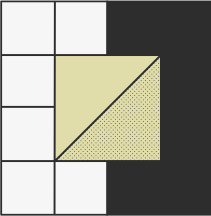
The system must have provisions to :

- **register with username, mobile no. and OTP .**
- **login with mobile no and OTP**
- **view prescriptions.**

Doctor:

The system must have provisions to :

- **register with username, mobile no. and OTP .**
- **login with mobile no and OTP**
- **create prescriptions for patients using username**
- **search patients prescriptions with user's mobile no. and OTP**



Requirements w.r.t Stake Holders

Pharmacist :

The system must have provisions to :

- **register with username and mobile no. and OTP .**
- **login with mobile no and OTP**
- **search patient's prescriptions with the user's mobile no. and OTP**

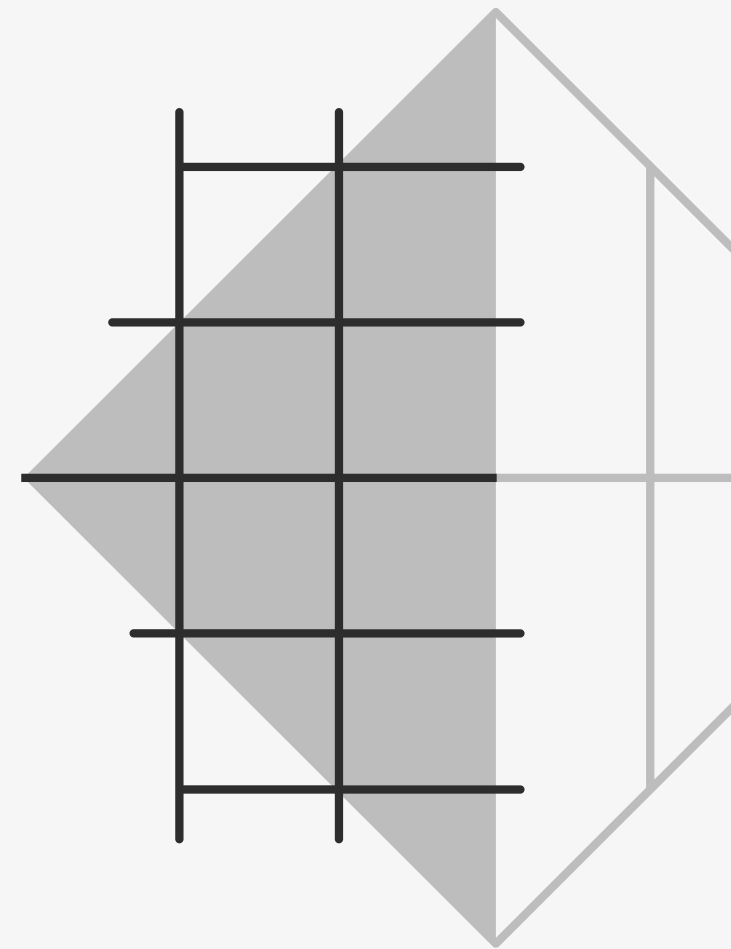
Admin :

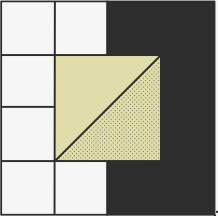
The system must have provisions to :

- **register with username and password**
- **login with username and password**
- **manage database and users**

Viewpoint

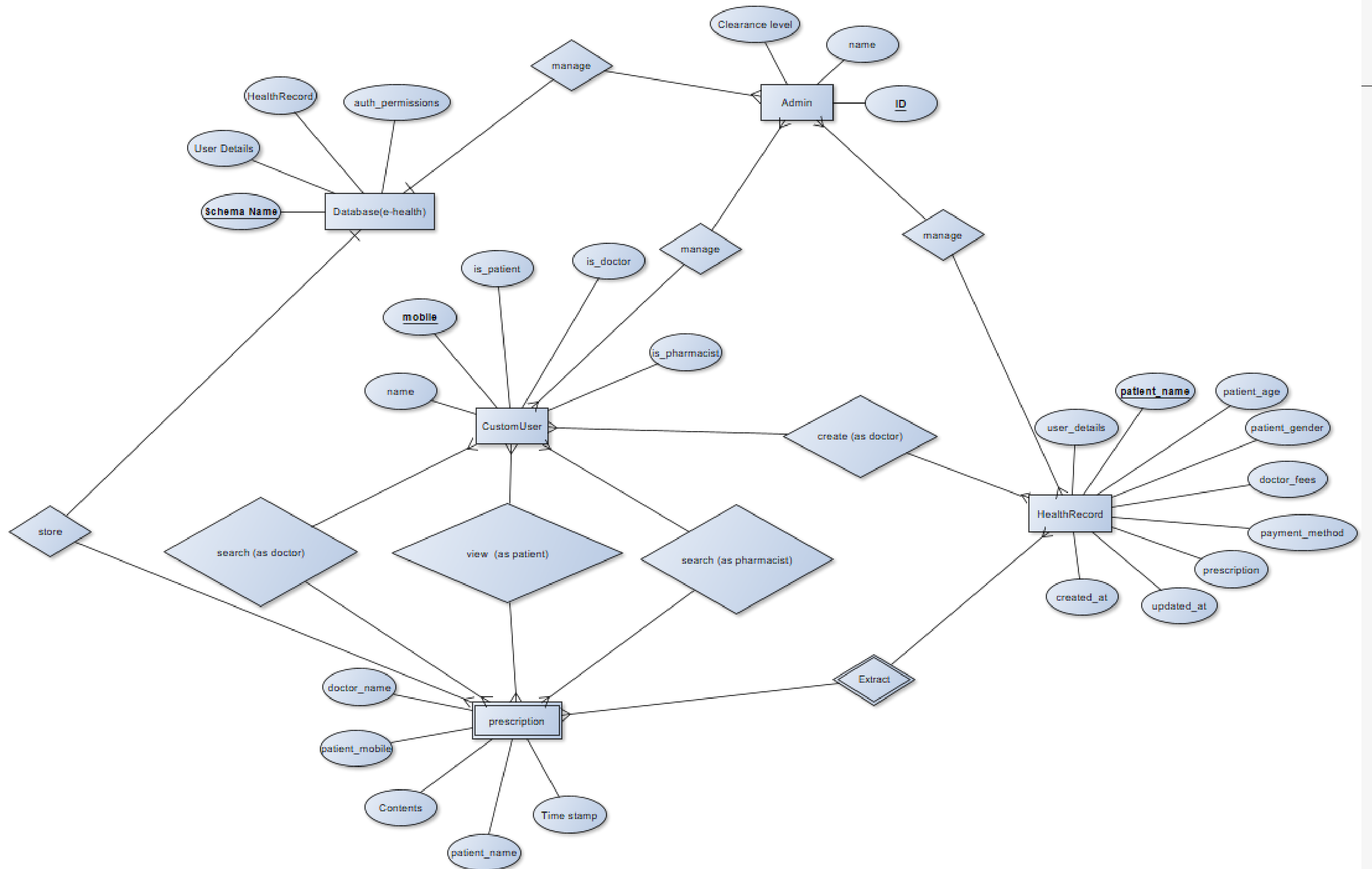
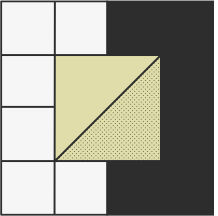
The Viewpoint selected is the Structural viewpoint. Structural Viewpoint involves handling large amounts of data which is all our system is about. The main functionality is for doctors to create prescriptions which can be stored in a centralized servers so that patients, doctors and pharmacists can access them in an instant. The functional viewpoint is not chosen as it is typically used to depict a lot of processes and data transformations which are not being done here. The behavioral viewpoint is a bad fit because there are no interactions taking place and all the actions performed by the system are not dependent on states.

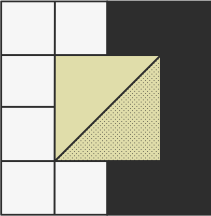




System Model

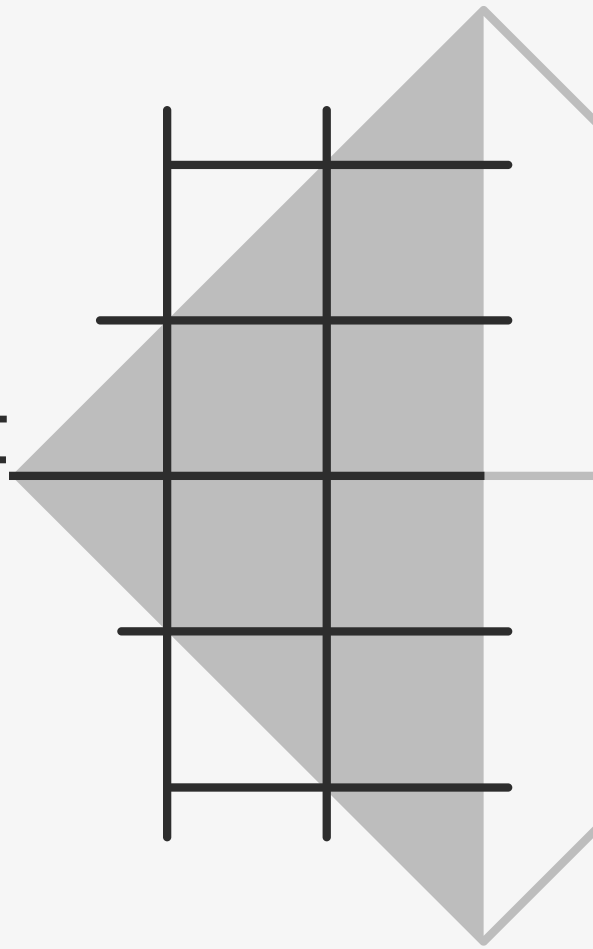
Since we have decided to use structural viewpoint. We have chosen Entity Relation model as a system model





Architecture

Client server architecture is used in our project as it is the most suitable. This involves storing data in a centralized server while doctors, patients and other stakeholders can directly access information from these servers.



Architecture

3 tier architecture

Tier 1:

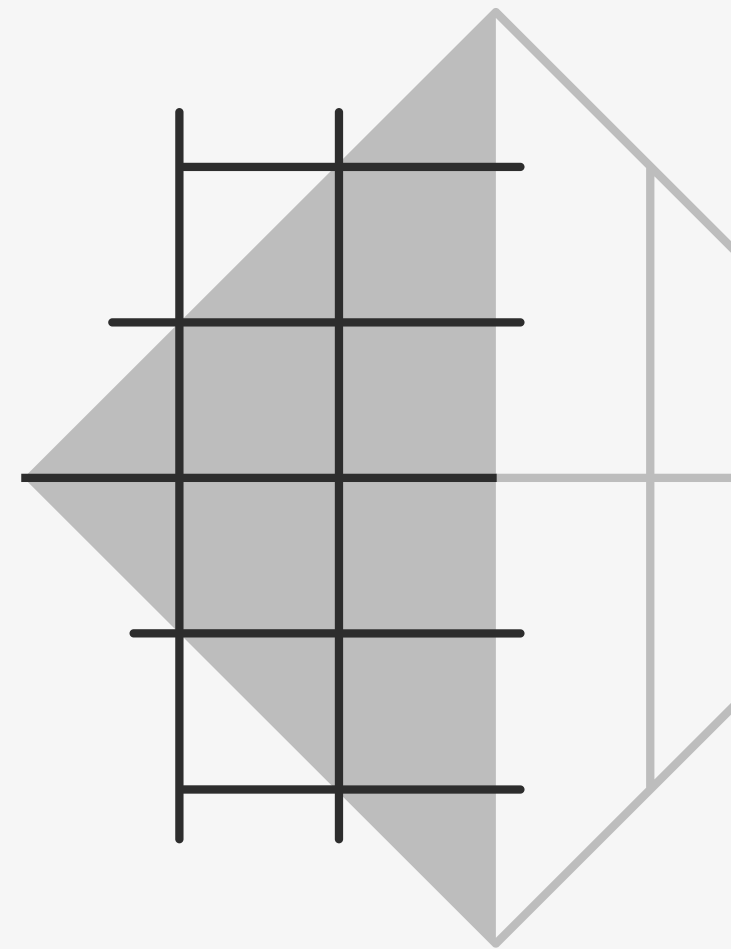
Stakeholders access web application through the clients (terminals and mobiles)

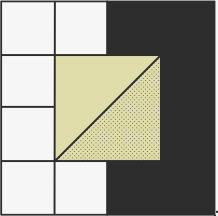
Tier 2:

Web Server which hosts One Health One Nation website

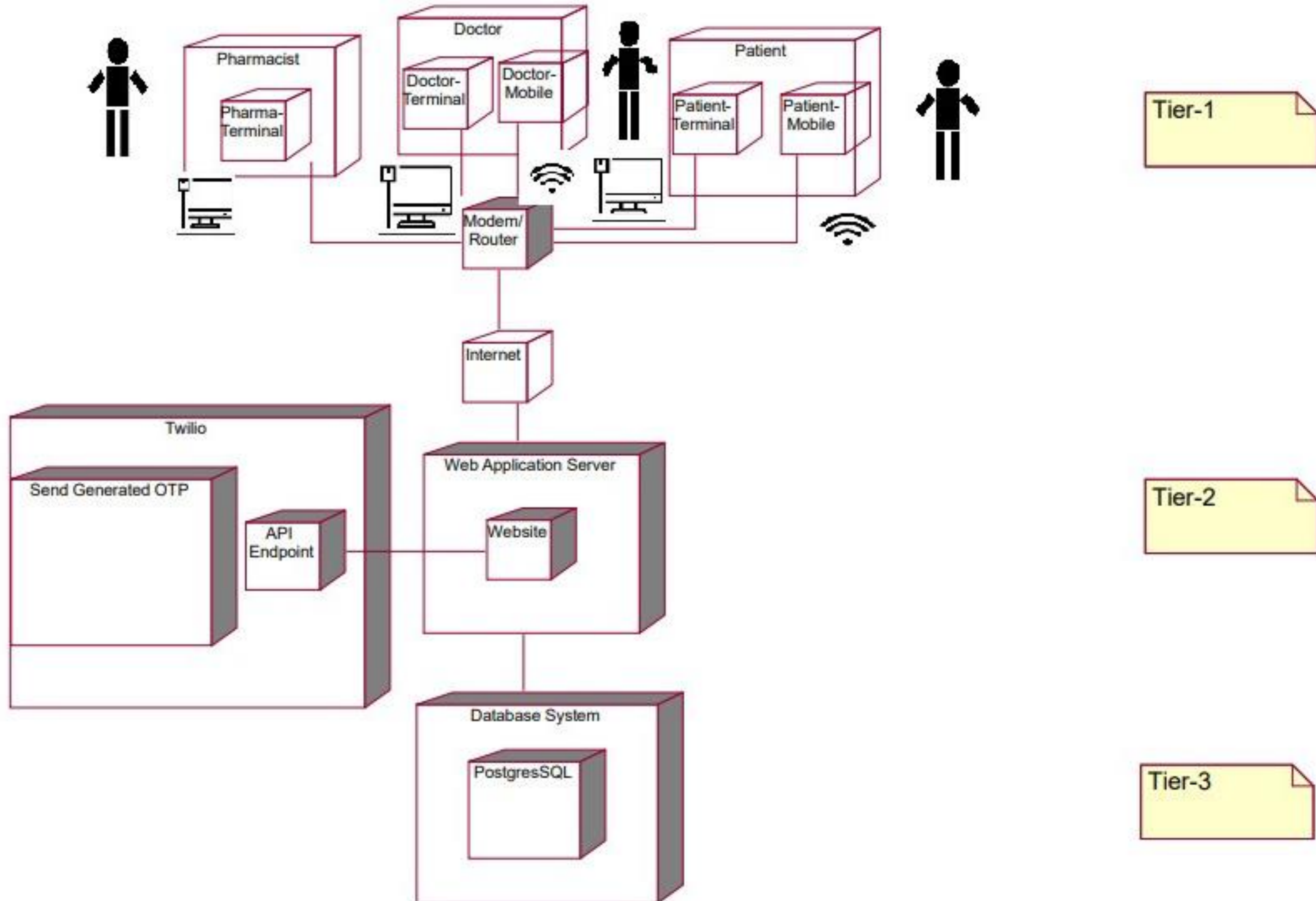
Tier 3:

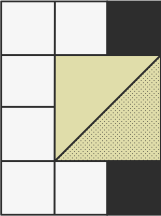
Database where prescriptions and user details are stored.





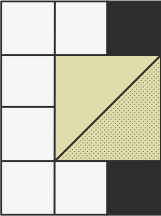
Deployment Model





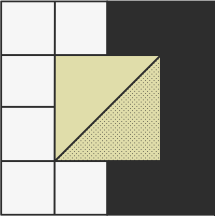
Test Cases

Test Id	Test Case	Test Data	Expected Outcome	Actual Outcome	Test Status
1.1	If (username not null)	“John Doe”	Logged in success	Logged in success	Pass
1.2	Else if(username null or username length is less than 15)	“”	Please Enter valid Username	Please Enter valid Username	Pass
2.1	If (Phone Number exist)	1234567890	Logged in success	Logged in success	Pass
2.2	Else If (Phone Number already exists)	1234567890	Phone Number Already Exists	Phone Number Already Exists	Pass
2.3	Else If (Phone Number is not null and length =10 and only numbers)	234567890	Logged in success	Logged in success	Pass



Test Cases

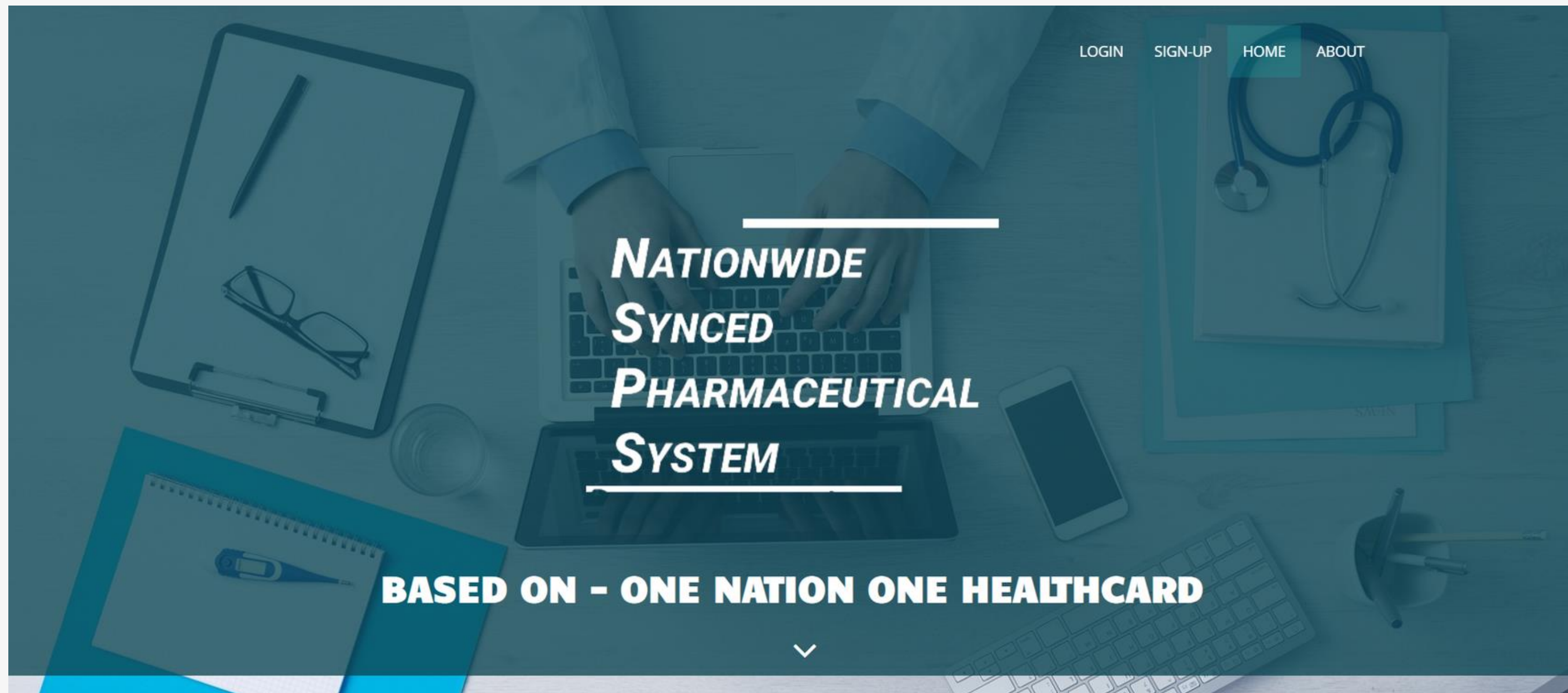
Test Id	Test Case	Test Data	Expected Outcome	Actual Outcome	Test Status
2.4	Else	12345	Invalid Phone Number	Invalid Phone Number	Pass
3.1	If (no checkbox selected)	Doctor	Alert(Select a role)	Alert(Select a role)	Pass
3.2	If (multiple checkboxes are selected)	Doctor ✓ Patient ✓	Alert(Select only one role)	Alert(Select only one role)	Pass
3.3	Else	Doctor ✓	Login Success	Login Success	Pass
4.1	If(Correct OTP)	Correct OTP(1234)	Login Success	Login Success	Pass



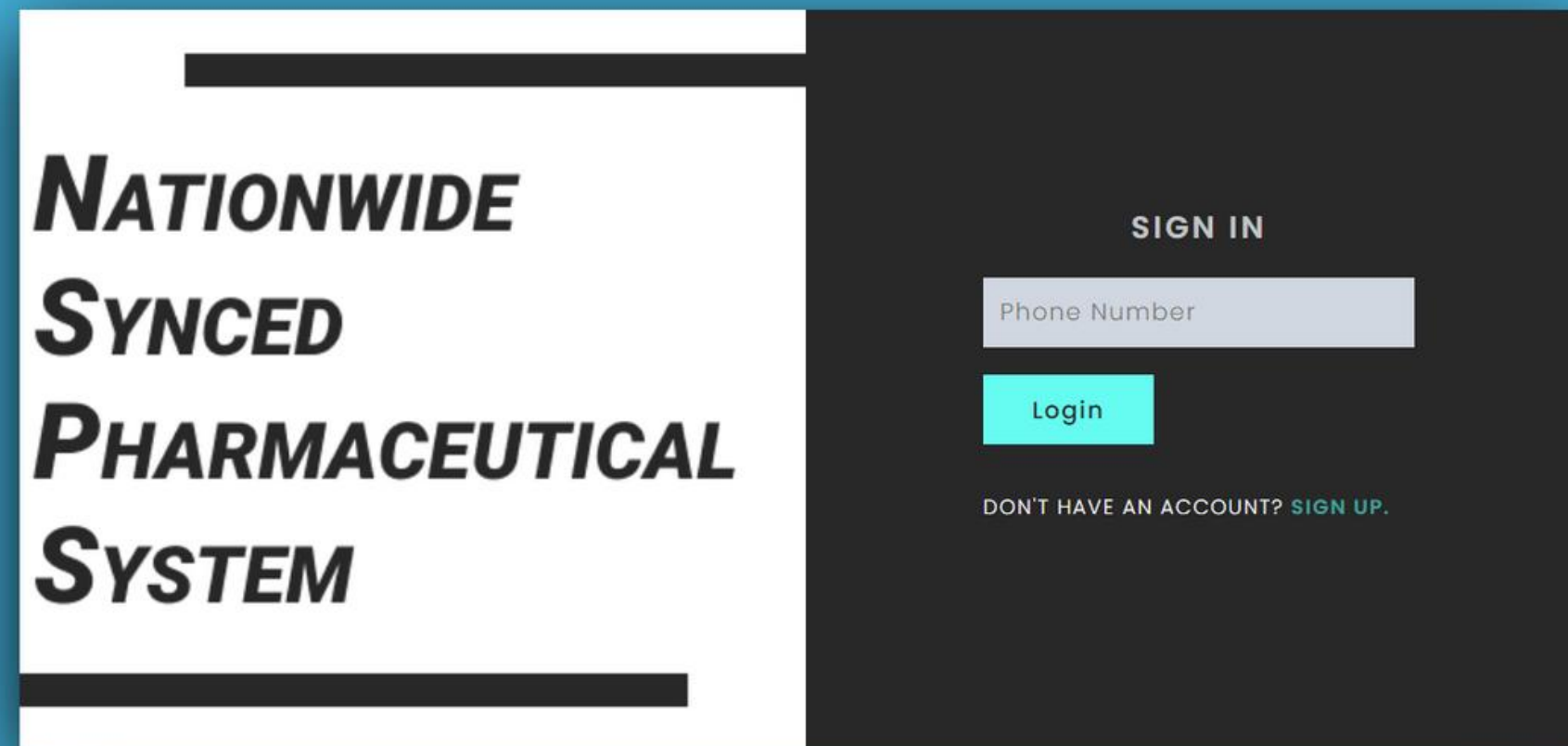
Test Cases

Test Id	Test Case	Test Data	Expected Outcome	Actual Outcome	Test Status
4.2	Else	Incorrect OTP(01)	Redirect to Login Page	Redirect to Login Page	Pass
5.1	If server Busy(status=503)	Login details	Logged in success	Server Busy	Fail
5.2	Else	Login details	Logged in success	Logged in success	Pass

User Interface Design



User Interface Design



The image shows a user interface design for a pharmaceutical system, presented as a mockup on a blue background. The interface is divided into two main sections: a white left panel and a dark grey right panel.

Left Panel (White):

- Contains the text **NATIONWIDE SYNCED PHARMACEUTICAL SYSTEM** in a bold, sans-serif font, arranged in four lines.
- Is framed by a thick black horizontal bar at the top and a thinner one at the bottom.

Right Panel (Dark Grey):

- Contains the text **SIGN IN** at the top.
- Features a light grey input field with the placeholder text "Phone Number".
- Includes a red "Login" button.
- At the bottom, it says "DON'T HAVE AN ACCOUNT? [SIGN UP.](#)" in a small, light grey font, with "SIGN UP." being a red link.

User Interface Design

The image displays a user interface design for a healthcare application, presented as a mockup on a blue background. The interface is divided into two main sections: a dark-themed registration form on the left and a white-themed branding area on the right.

Registration Form (Left):

- Title:** CREATE AN ACCOUNT
- Fields:** Two light blue input fields for "Username" and "Phone Number".
- Role Selection:** Three radio buttons labeled "Patient", "Doctor", and "Pharmacist".
- Action:** A red "Sign up" button.
- Link:** A link "ALREADY HAVE AN ACCOUNT? SIGN IN." in red text.

Branding Area (Right):

- Top Section:** The text "ONE NATION" in a large, bold, black sans-serif font, with a large black number "1" to its right. A thick black horizontal line is positioned below the text.
- Bottom Section:** The text "ONE HEALTHCARD" in a large, bold, black sans-serif font.

User Interface Design

OTP VERIFICATION

OTP expires in 60 seconds.

Submit

User Interface Design

[DR XQC](#) [LOGOUT](#) [ABOUT](#)

DOCTOR'S PORTAL

Find Patient's Record

Patient Phone Number

Search Prescription

Create Prescription

User Interface Design

[HOME](#) [DR.XQC](#) [LOGOUT](#) [ABOUT](#)

DOCTOR'S PORTAL

Add Prescription

Your Registered Name

Your Age

Gender

Doctor Fees

Payment Method

Details and Prescription

SUBMIT

User Interface Design

DR XQC LOGOUT ABOUT

DOCTOR'S PORTAL

Patient Records

Patient's Name	Doctor fees	Age	Prescription	payment method	Date
abc	1234	20	trial 1	Cash	Nov. 7, 2022, 8:32 p.m.
abc	1234	20	trial 2	Cash	Nov. 7, 2022, 8:32 p.m.
abc	12423515	20	test 3	Cash	Nov. 9, 2022, 6:05 p.m.

Creators :

Software Engineering
Group-4
J-Component

Quick Links

- Home
- Create prescription

User Interface Design

PATIENT ABC LOGOUT ABOUT

PATIENT'S PORTAL

Personal Records

Patient's Name	Doctor fees	Age	Prescription	payment method	Date
abc	1234	20	trial 1	Cash	Nov. 7, 2022, 8:32 p.m.
abc	1234	20	trial 2	Cash	Nov. 7, 2022, 8:32 p.m.
abc	12423515	20	test 3	Cash	Nov. 9, 2022, 6:05 p.m.

Creators :

Shreyas bhardwaj
Garv Mittal
Harshanbarath
Daniel jebarson
Ruiula Nitin Patil

Quick Links

- Home

Thank You

Software Requirements Specification

for

One Nation- One Health

Version 1.0 approved

Prepared by:

Garv Mittal

Shreyas Bhardwaj

Daniel J aberson

RS Harshanbarath

Rujula Nitin Patil

Org name: GSRDH

6-11-2022

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1. Introduction

1.1 Purpose

This project will be implemented on a network such that entire population of India can use it easily. A cloud service such as suppliers like Amazon Web Services (AWS). will host the application. The scope of the project is to provide an interface to the users that are the patients and doctors. The doctors can access the history (of patient), if the patient requests. They can view the records and provide prescriptions.

1.2 Document Conventions

It is recommended to read the complete SRS document in order, as given in the index. Each and every paragraph in the document provides us with a new view point/ explanation about the product. Like scope gives the initial things about the project including aim, objective and application, etc .

1.3 Intended Audience and Reading Suggestions

This document is intended for all kinds of user whether it be a newbie in the industry. And also, for developers and project managers to get the basic idea. This SRS document contains all the necessary stuff required to get started with the project, including aim scope references System features, functional and nonfunctional requirements. It is organized in proper IEEE format, which makes it easier to locate the sections and sub sections. The sections that the user must go through to get the idea are problem statement, scope and the conclusion.

1.4 Product Scope

AIM:

To develop a software that creates a nationwide synced pharmaceutical system to enable doctors to create and track prescriptions of patients across India.

Objective:

- Efficiency
- Optimized
- Maintainable
- Reliability

1.5 References

<<https://ebooks.iospress.nl/doi/10.3233/978-1-60750-870-0-3>

Authors T. Takahashi, Y. Sekita, S.H. Mandil

Johar, M. (2009). The impact of the Indonesian health card program: A matching estimator approach. *Journal of Health Economics*, 28(1), 35-53.
<https://doi.org/10.1016/j.jhealeco.2008.10.001>

Authors [s](#)

Year of publish : January 2009

2. Overall Description

2.1 Product Perspective

This system is a unique one. It helps provide patients a comfortable and hassle-free method of contacting doctors, receiving prescriptions and ordering medicines with maximum privacy for their data. Doctors can access the past records of a patient with the consent of the respective patient and help provide the patient with a quick and accurate diagnosis of the problem. Patients/customers don't have to worry about storing their prescriptions as this system stores the prescription for them online allowing it to be accessible at any time through a device with a browser.

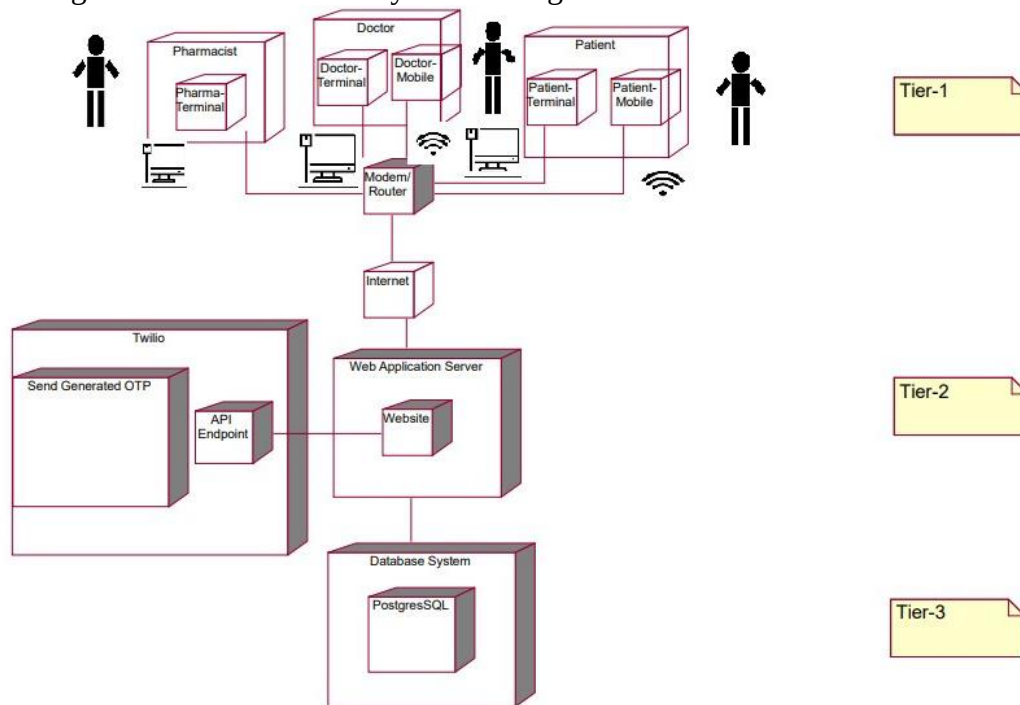


Figure 1: Deployment Model

2.2 Product Functions

- To provide the patients a comfortable and hassle-free method of contacting doctors, receiving prescriptions and ordering medicines.

- Patients don't have to store their prescriptions as this system stores the prescription for them, allowing them to be accessible online at any time through a device with a browser.
- Doctors can access the past records of a patient with the consent of the respective patient and help provide the patient with a quick and accurate diagnosis of the problem.

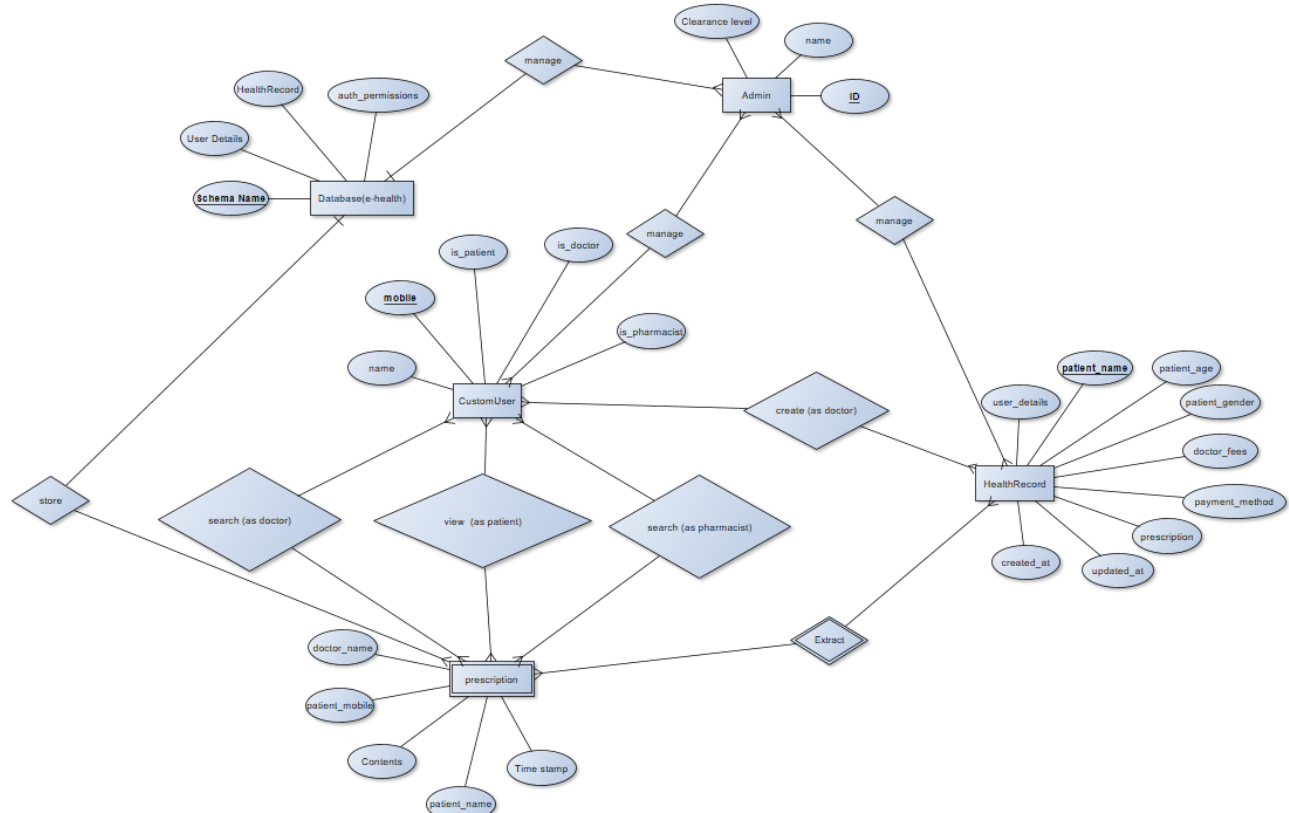


Figure 2: Entity Relationship Diagram

2.3 User Classes and Characteristics

The various user classes include:

- Patients: The system must have provisions to let patients register, login and view prescriptions.
- Doctor: The system must allow doctors to register, login, create prescriptions for patients using username and search patients' prescriptions.
- Pharmacist: The system must have provisions to allow pharmacists to register, login and search patient's prescriptions.

2.4 Operating Environment

Patients, doctors and pharmacists can access prescriptions and any point of time through any device with a browser.

2.5 Design and Implementation Constraints

This project is hosted locally and needs to be scaled up, so that it can be used throughout the nation. Another issue is that this project is only a web application and can be converted into a mobile app in the future. The quantity of data that needs to be stored in the databases is very large and will only keep increasing.

2.6 Assumptions and Dependencies

Twilio is a paid application which allows OTPs to be sent, allowing us to fully experience the system with no roadblocks.

3. External Interface Requirements

3.1 User Interfaces

The users use the website to communicate with the database to store and retrieve information. The system automatically generates OTPs for verification using an API call and sends it to the user's mobile

OTP Verification interaction between user and system + system and API.

3.2 Hardware Interfaces

The users use Terminals and mobile devices to connect to the internet through the modem. The server stores the web application which is in turn connected to the database.

The most important protocol used is the HTTPS (Hyper Text Transfer Protocol Secure) for web services and communication using http request and responses.

3.3 Software Interfaces

The product uses PostgreSQL as database management tool as it is a relational DBMS and is perfect for the product. The product uses Twilio API to send OTPs from a unique number to the users. Below are the softwares versions used to build this product.

asgiref==3.5.2

certifi==2022.9.24

charset-normalizer==2.1.1

Django==4.1.2

django-extensions==3.2.1

idna==3.4

jsonfield==3.1.0

PyJWT==2.6.0

pyotp==2.7.0

pytz==2022.5

requests==2.28.1

sqlparse==0.4.3

twilio==7.15.0

tzdata==2022.5

urllib3==1.26.12

whitenoise==6.2.0

3.4 Communications Interfaces

The communication functions used in the project were for sending OTPs only and this is done with an API call to Twilio application. The messaging is done using HTTP communication standards. On the OTP being delayed, it is resent and if the OTP is delayed in being entered, it is rejected. The OTP message usually arrives in 5 seconds at max.

4. System Features

4.1 Role Selection:

There are different roles for different people like doctors, patients, etc. Each role will have limited access depending on what they need.

4.1.1 Doctor:

The doctor can see the patients who are under him and can create prescriptions, filter by number, etc. They can also see the health history of the patients.

4.1.2 Patient:

Patients can see the prescription described by the doctor to themselves and can pay the fees either as cash or by card depending on which method they want to pay.

4.1.3 Pharmacist:

There is also another role called pharmacist who can give the tablets according to the doctor's description to the patients.

4.2 OTP System:

There is an OTP system in this project which prevents other people from logging into the respective user's account and thereby aiding security. With this, the user can keep track of who is seeing their profile and can prevent impersonation of the users. In such a way this system provides security and keeps the user's information private.

4.3 User-Friendly System:

One Nation One HealthCare is mainly dependent on the user's experience. The users can easily see their prescriptions and buy tablets from the pharmacist. The user can edit their profile as per their wish but they need to verify for other roles such as doctors, pharmacists, or admin. If the user faces any issue, they can contact the support team who will help to fix the user's problem. To avoid losing patients building a friendly system is very important so we made a user-friendly system that satisfies the user's needs.

5. Other Nonfunctional Requirements

5.1 Software Quality Attributes

- a) **Availability:** The users should have data connectivity or get access to the Wi-Fi if they have basic internet connection or nearby.
- b) **Reliability:** The software is reliable and unforeseen calamities like power shortage or system crash will not do harm to your profile or undergoing cash transactions. They will simply be put on hold.
- c) **Portability:** The software is basically built for both mobile and PC
- d) **Maintainability:** Software updates will be made available every month to maintain smooth running.
- e) **Security:** The system is highly secure. All confidential information of user in their accounts are hidden from other parties up to required extent.
- f) **Modifiability:** Application is not open source and hence cannot be modified without developer consent.
- g) **Safety requirements:** There are no safety requirements with this application. In case of device hazards, data flow is stopped and reverted to previous safe state hence not corrupted or compromised.
- h) **Flexibility:** Application is easily modifiable by developer to maintain and update with changing environment

5.2 Performance Requirements

- a) **Response time:** The system shall give response within 1 second after giving patient/doctor information
- b) **Capacity:** System shall support 2000 records at a time
- c) **User Interface:** The UI screen shall respond within 5 seconds
- d) **Scalability-Horizontally scalable:**

- e) Centralization: Centralized systems are systems that use client/server architecture where one or more client nodes are directly connected to a central server. Here, a client sends a request to a company server and receives the response

5.3 Security Requirements

The system shall protect the data and services from unauthorized access. The system shall also provide authentication and secure transaction. System should provide highest possible security mechanism in order to protect critical information. System should run privately over public network. System should restrict all the non-member of the project to get access to the system. The system shall provide a mechanism of user authentication to unambiguously identify a user

6. Other Requirements

Internationalization can be done with our project to be made available across all platforms and ranges. We can do horizontal scaling if there is a need to spread the product across countries. Another step that can be taken is to convert code into java so that it becomes platform independent.

Legal

- National Digital Health Mission

- Personal health data

- Personal Data Protection Bill 2019 adding anonymity to stored health records